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CS 193: Independent Study

Overall

This independent study class, under the supervision of Dr. Tamar Shinar, is the application of computer graphics particularly in animation using Autodesk Maya. In essence, computer graphics is the combination of art and computer science. This class aims to create an animated short by following the general animation pipeline, exposing myself to 3D computer graphic art, and learning the technical applications of computer graphics in animation.

Week 1: Concept

Storytelling was the main focus for this week. I gathered all my information from KhanAcademy's Pixar in a Box where they taught me how to incorporate my life into a story. A personal anecdote makes the story honest and it becomes very vivid when emotions become involved. Identify some emotion that connects you to the story and allow to express yourself in the story. Creating a story becomes easier when it is from real human emotions and experiences.

Week 2: 1-2 Concept Art / Script / Storyboard

This week, I finalized a script from a list of story concepts I created. For my story, I chose to incorporate a fun character named Pómo who becomes impatient with his routine and wants to speed up his life which results in many consequences. The name Pómo comes from the Greek word ανυπόμονος which is pronounced as anypómonos. I shortened it to Pómo. The setting is within a clock enclosure where Pómo, his bed, office, and lunch table live on the hour hand. Attached is the script for *Pómo*.

Pómo

Supervising Instructor: Dr. Tamar Shinar

Created by Jonathan Kaneshiro

Fade into overview of watch enclosure

Day 1:

(Overhead shot [Parallel to ground])

Zoom into Pómo sleeping on his bed

Clock ticks to 6 am (entire ground flashes in light green)

Character wakes up (groggy)

(Overhead shot)

Hour hand moves to 7 am (Animation: hour hand moving to 7)

Zoom in to name tag pomo

Other character's hand moves in with papers (data entry job)

(Overhead shot)

Hour hand moves to 12pm (Animation: hour hand moving to 12)

Other character's hand gives him a sandwich (lunch)

Sandwich bite

(Overhead shot)

Hour hand moves to 8pm (Animation: hour hand moving to 8)

Jumps to bed

Day 2:

Clock ticks to 6 am (shakes the whole "room/bed")

Character wakes up (even more groggy!!)

(Overhead shot)

Hour hand moves to 7 am (Animation: hour hand moving to 7)

Other character's hand moves in with papers (data entry job)

(Overhead shot)

Hour hand moves to 12pm (Animation: hour hand moving to 12)
Other character's hand gives him a sandwich (lunch)
Sandwich bite

(Overhead shot)

Hour hand moves to 8pm (Animation: hour hand moving to 6)
Goes to bed

Day 3:

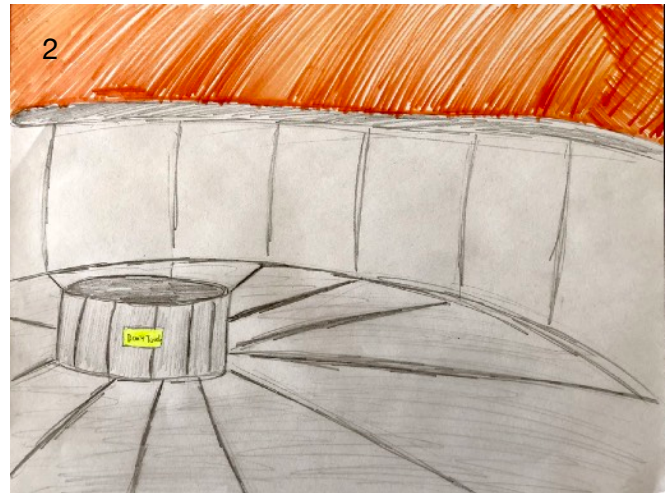
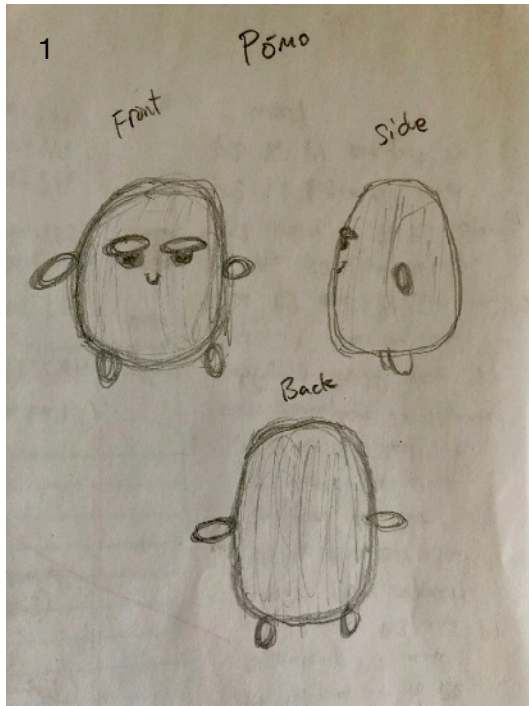
Clock ticks to 6 am (shakes the whole "room/bed")
Character is misshapen and slowly gets out of bed
Looks at the center of the watch and notices a Do not touch
button

Zooms in to the button (don't press - > (will go faster))
Pomo ecstatically presses it
The "conveyor belt" speeds up

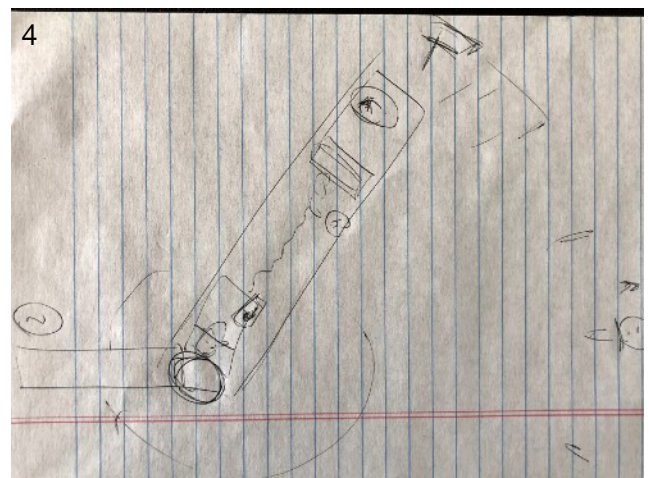
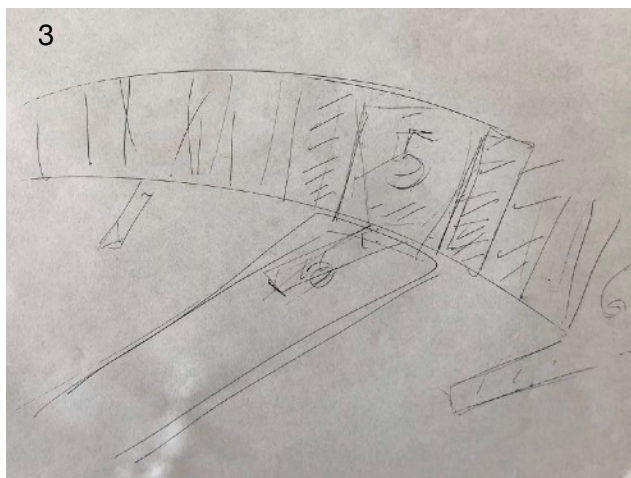
Pomo tries to keep up by jogging/ running faster
Gets hit in the face with the sandwich
Gets hit in the face with papers as the conveyor belt keeps
going faster
Knocks over his bed
Tries to reach for the button to stop it (dramatic)
Successfully presses it
Big sigh

Week 3: 1-2 Concept Art / Script / Storyboard (Continued)

Concept art was fairly new to me as I am an amateur when it comes to drawing. Nonetheless, I imagined the character to have a very deformable body with rounded edges. I drew a rough storyboard illustrating each scene of Pómo's story.

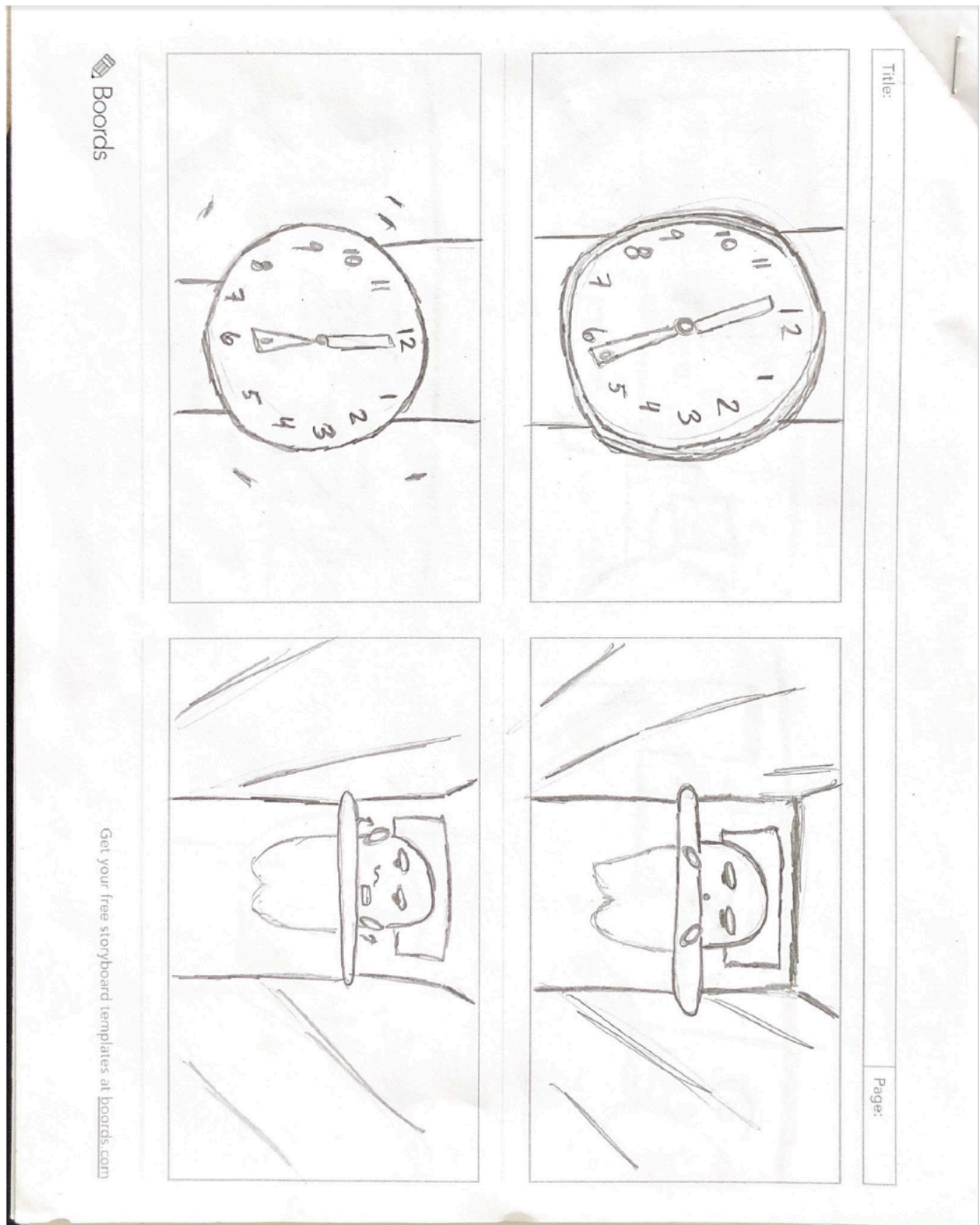


1. Concept art of Pómo
2. First concept art of the clock
(Envisioned it as a conveyor belt but changed it to an actual clock with clock hands and numbers)



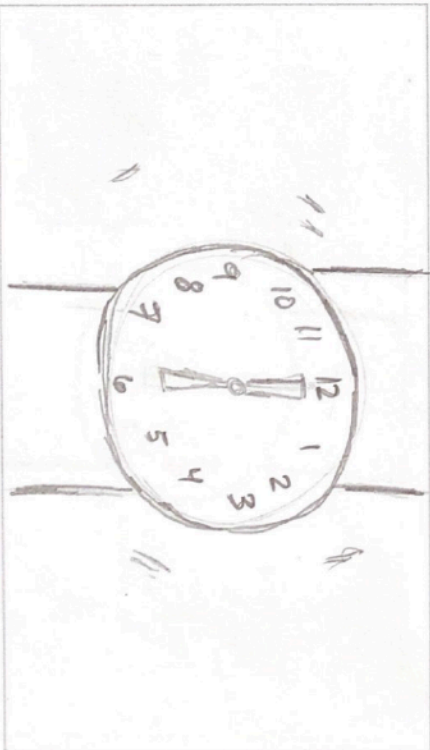
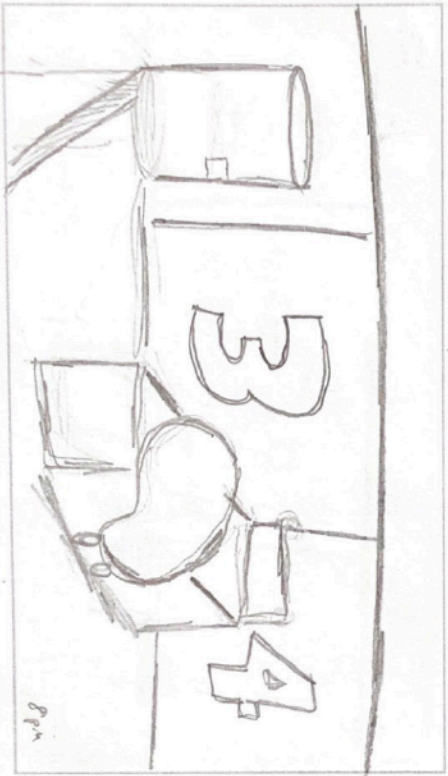
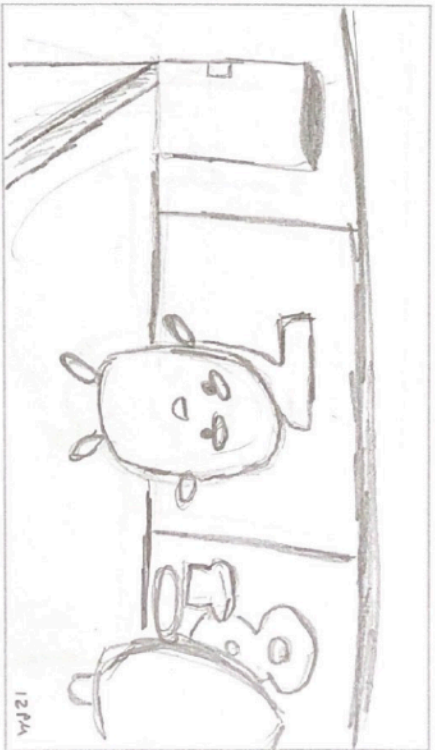
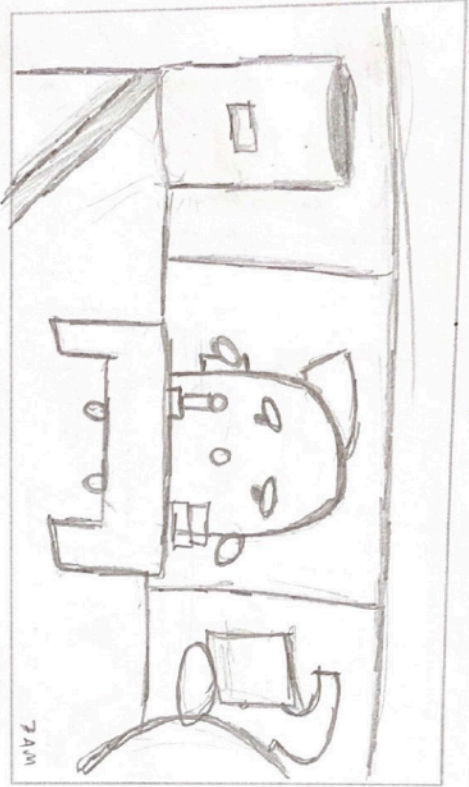
3. Adjusted concept art of the clock incorporating the conveyor belt appearance (Shinar)
4. Noting various camera positions. This shows the perpendicular camera angle. (Shinar)

The following pictures depict the storyboard.



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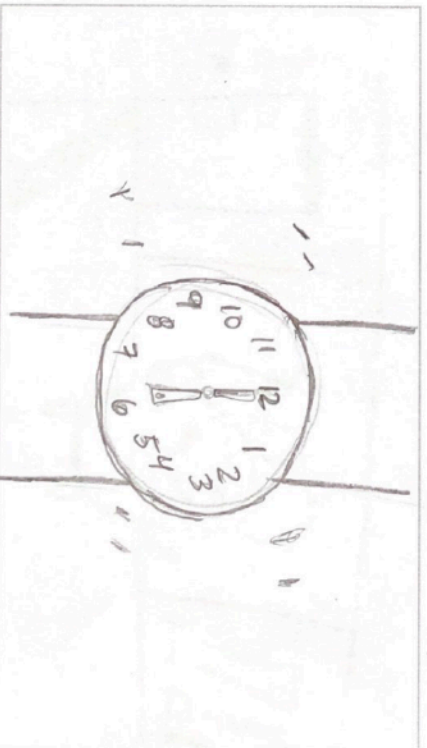
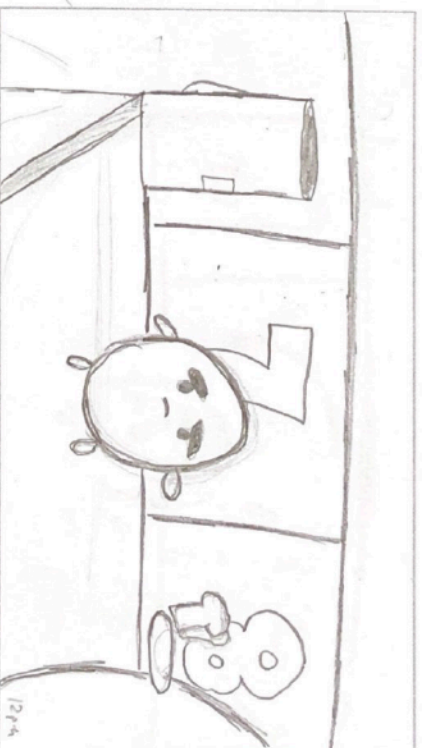
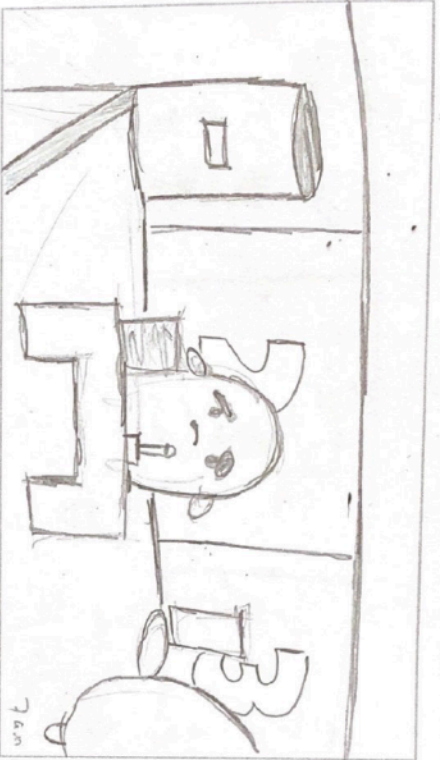


Boards

Get your free storyboard templates at boards.com

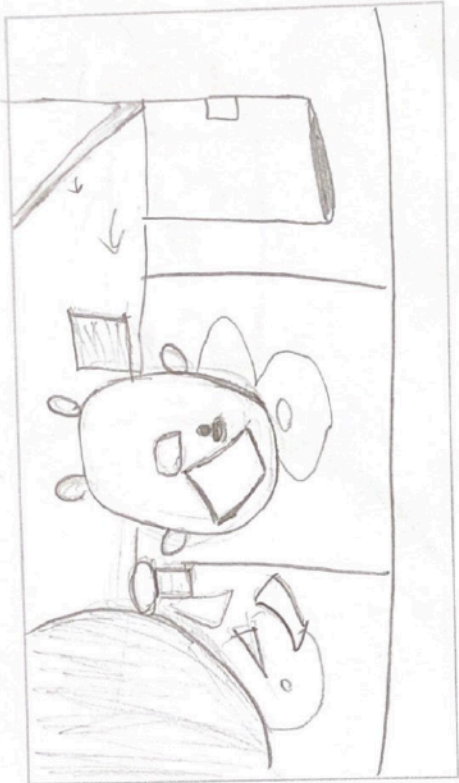
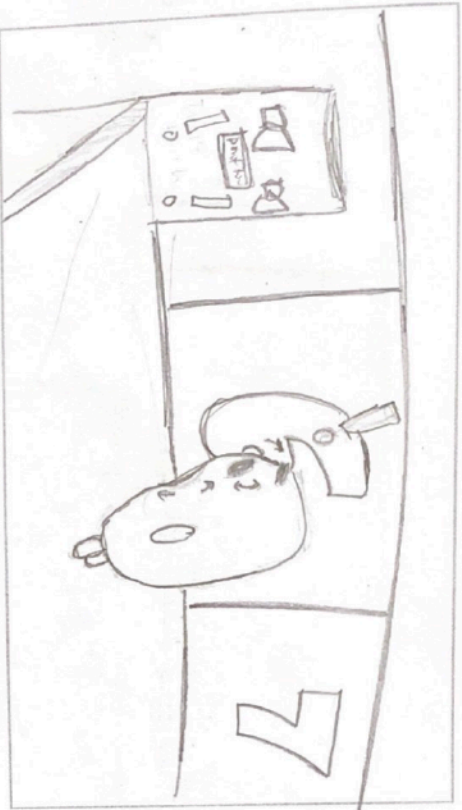
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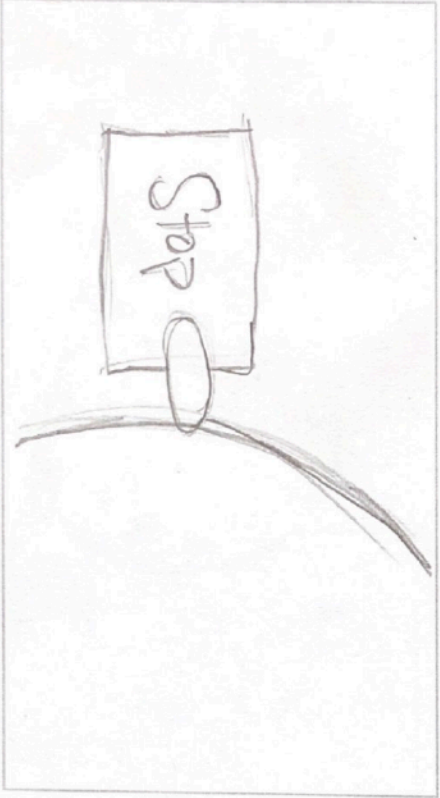
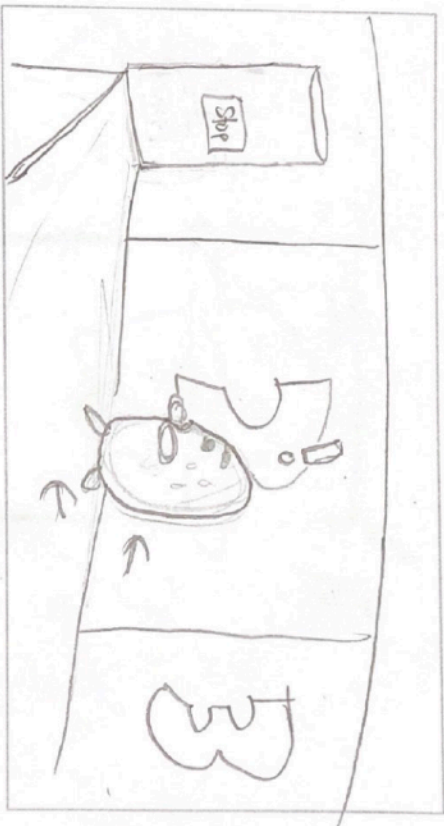
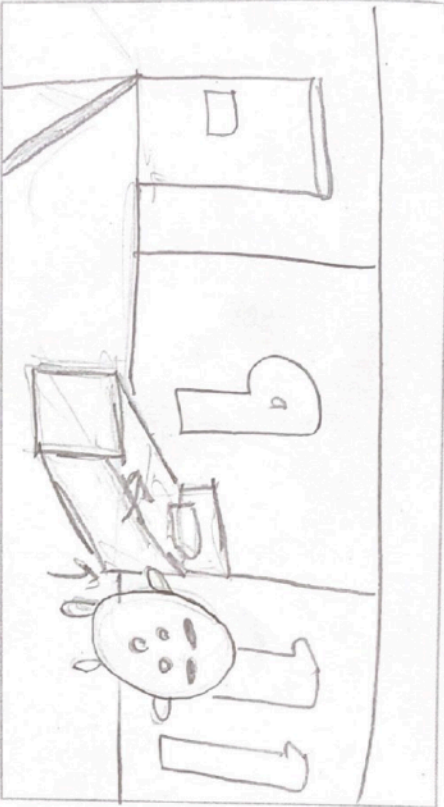
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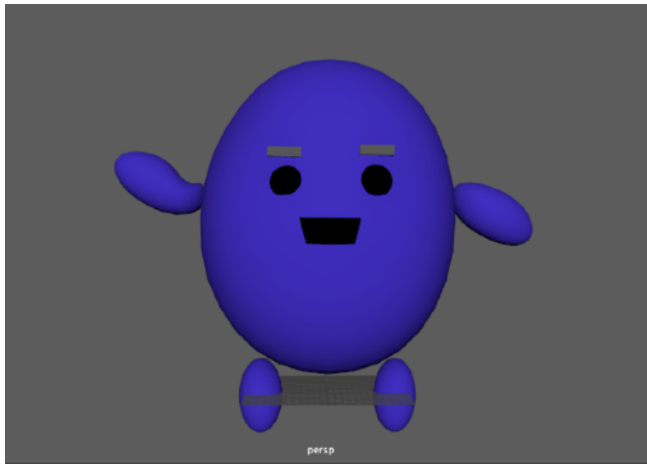
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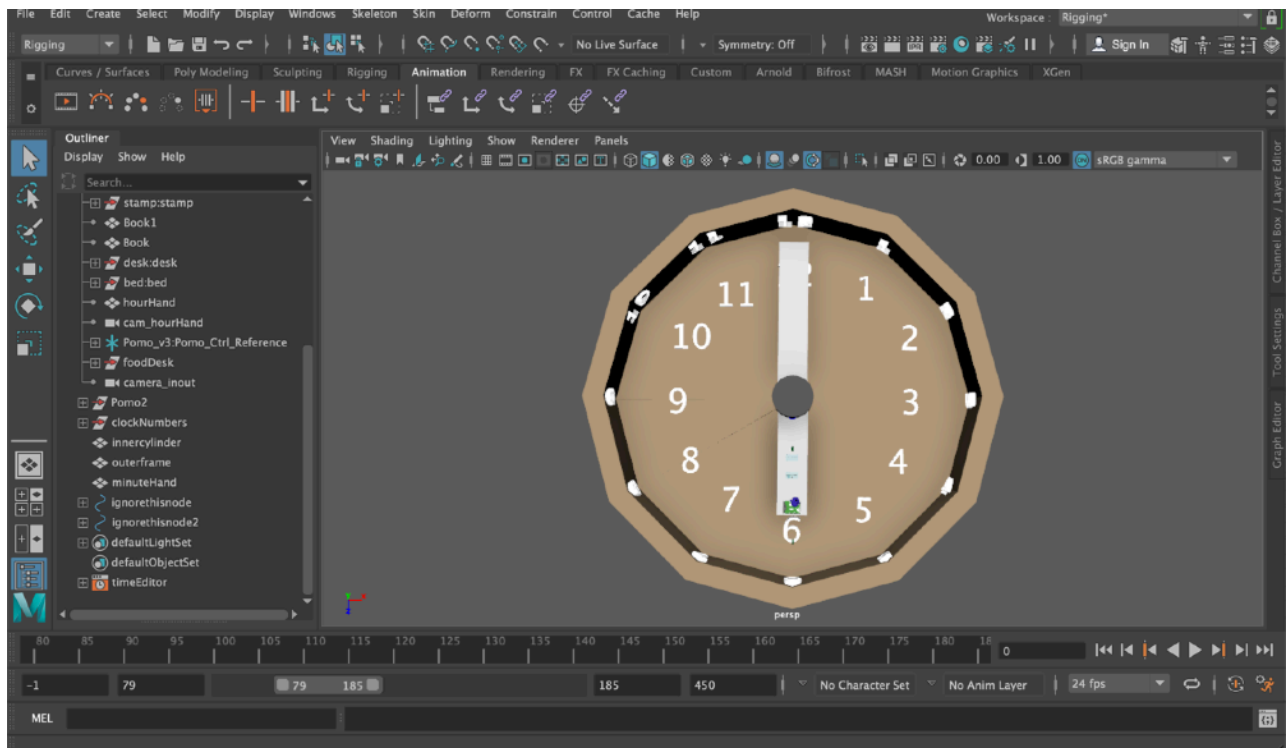


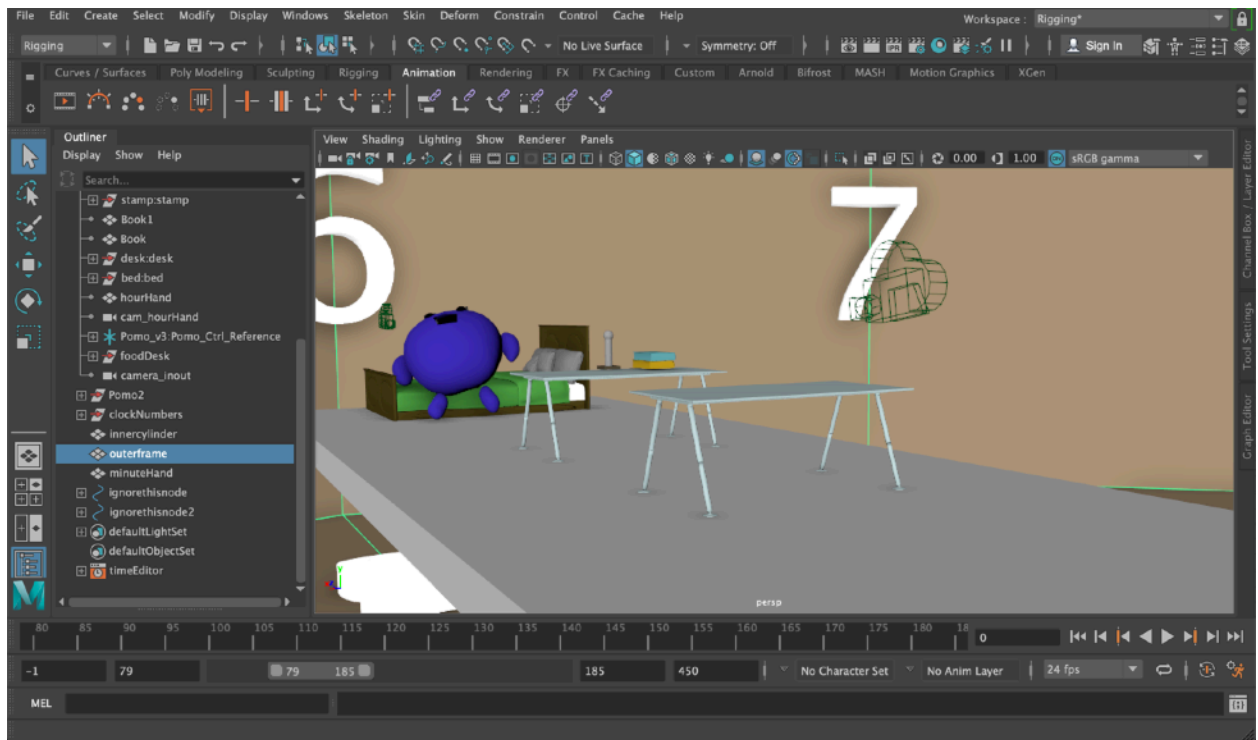
Week 4: 3D Modeling

I followed tutorials on how to create 3D models using Skillshare. I wanted my models to have fairly basic geometries so rendering would not be an issue in the future. The polygon count for most of my models were manageable for my GPU and CPU configurations. I incorporated bumps maps creating the illusion of depth into the models. Ambient occlusion was also in effect when integrating the bumps maps and various lighting conditions.



A basic geometric 3D model of Pómo
(Below) A perpendicular shot of the clock environment

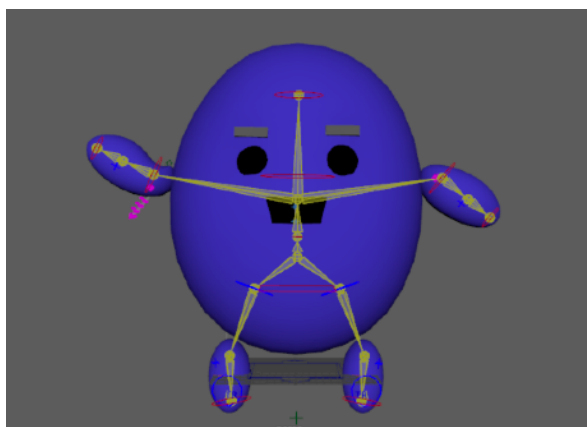




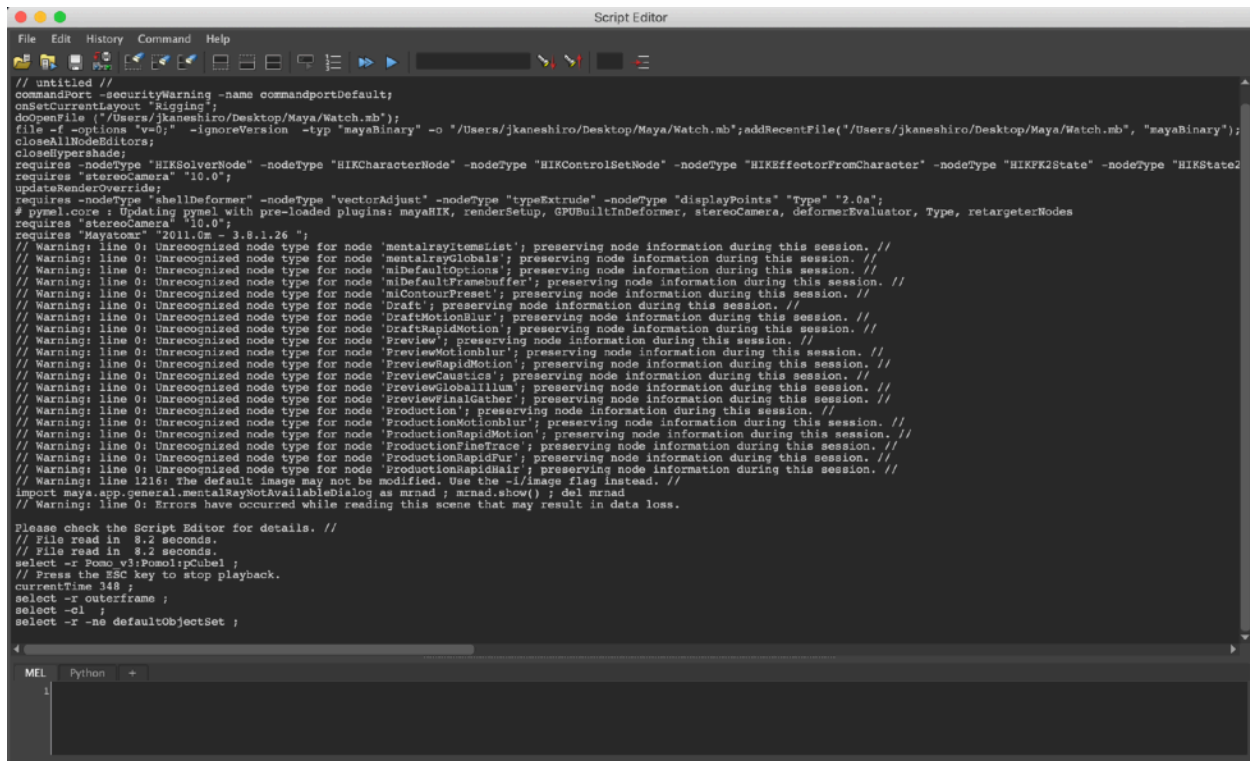
(Frame 0: Another camera perspective showing the geometries of Pómo on his bed, his desks, and environment.)

Week 5: Rigging

Rigging is the process of creating a skeleton for a 3D model. I first started out by editing a Maya script using the Maya Embedded Language (MEL) which involves slicing the body into individual proxies that make up the body of Pómo. However, Pómo does not have a similar body frame as an average human so I did not use the Maya script. Fortunately, Maya has a Quick Rig tool that adds a rig to a biped character. I manually positioned the guide joints to Pómo and then afterwards, a rig was created.



A skeleton-rigged Pómo with all his joints defined



(Example of Maya's script editor displaying warnings in MEL)

Week 6: Rigging (Continued) / Skinning

The definition of skinning is binding the skeleton to a 3D model allowing it to move. There are still skinning issues on Pómo where it needs to be refined but it is fairly sufficient at this time. To fix this, apply skinning weights to his entire body. The central control node is important as it controls the rigged skeleton as well as the polygon meshes encompassing the character's body. A hierarchy of body structures helps organize the connections between joints allowing for inverse kinematics to occur.

Week 7: Animatic

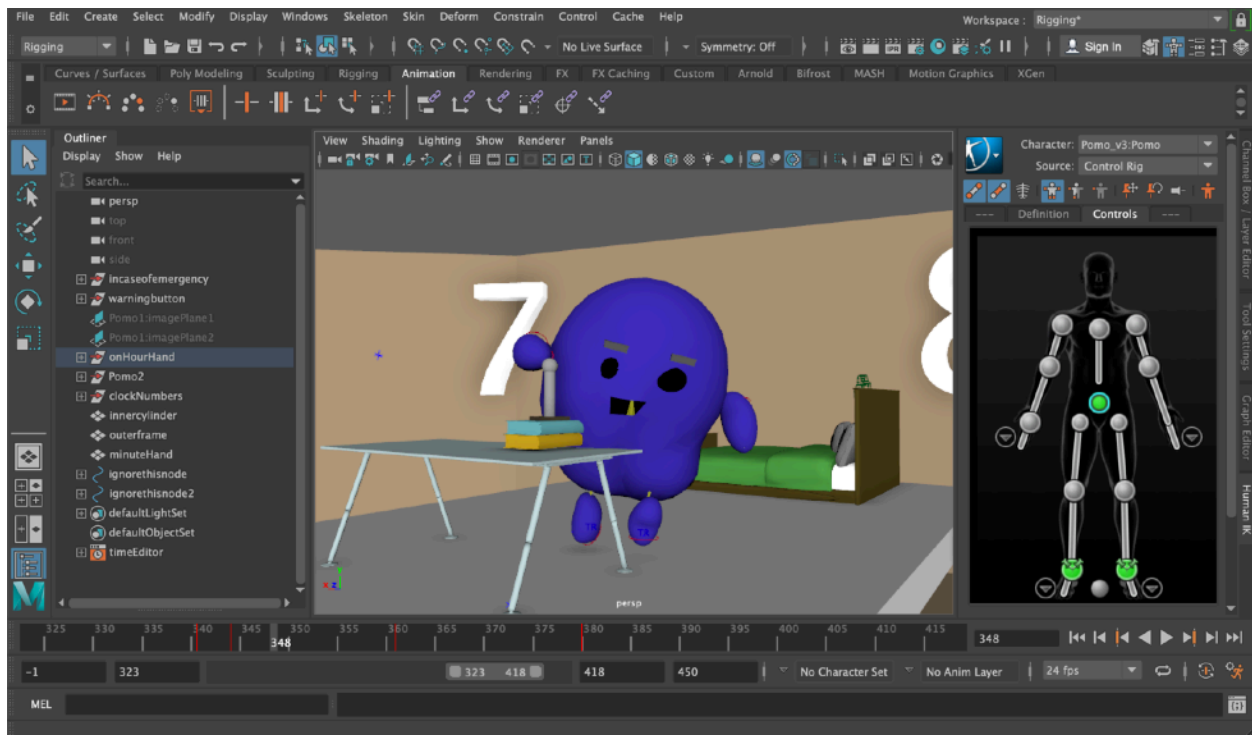
An animatic is a preliminary version of a movie containing consecutive sections of a storyboard where timing and camera perspectives are emphasized. I created an animatic for this week where I enhanced the storyboard with more visual aids such as darkening the lines and adding emphasis to key points in the story. For my animatic, I scanned the storyboard drawings using my iPhone and placed them into iMovie to ascertain the timing of each shot.

Week 8: Animation

For this project, I am focusing on getting the fundamental motions down. Scene one introduces Pómo waking up from his sleep, stretching, and then getting out of his bed. I am having trouble with laying him flat on his bed as his body and bed do not have a sense of weight or rigidness where physics are applied. This can be fixed later by incorporating physics-based simulation to his body and blanket. For each movement, I am adjusting the rotations and translations of Pómo and the hour hand and saving it as a keyframe. All the items on the hour hand including Pómo are translating and rotating in local space as all their movements are relative to the hour hand.

Week 9: Animation (Continued)

Scene two is Pómo walking to his desk, stamping his books, and eating his lunch. Unfortunately, I did not have enough time to finish animating him at lunchtime but I definitely will continue this after the quarter. I tried implementing some form of the 12 principles of animation into my project. One principle is squashing and stretching. This occurs when Pómo walks as it gives him a sense of weight and flexibility. I used Maya's HumanIK for all the controls for animation which uses inverse kinematics (IK).



(Frame 348: Figure of HumanIK along with Pómo reaching for his stamp)

Forward kinematics (FK) can be extra tedious during animation as animators have to directly change each joints' angles and positions. HumanIK allows users to change to IK or FK depending on the scene.

Week 10: Closing Remarks

I used Maya's standard renderer using the ffmpeg framework to create the movie file. It was easier to create a simple movie using ffmpeg than using a high-end video editing software.

I want to thank Dr. Shinar for supervising this class. Learning the animation pipeline was rewarding as I admire the work of animators, modelers, and riggers even more. Throughout this class, I struggled with 3D modeling as I do not have a strong artistic background. However, I still want to enhance my 3d modeling skills after this project. I really enjoyed creating different storylines as I delved into past memories to figure out a story. If I were to do this again, I would choose a storyline that elicits a more personal story as I would connect more with the story along with better conveying my life through animation.